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Course Name: 5CS018_Cybersecurity Architecture and Operations

Task 1

a) What are the common types of data security incidents reported?

- Common types of data security incidents reported are:

1. Data emailed to incorrect recipient
2. Phishing
3. Unauthorized access
4. Ransomware
5. Loss/theft of paperwork or data left in insecure location.

b) What are the 3 main cyber security incidents reported in these sectors?

i) Health:

1. Phishing
2. Ransomware
3. Unauthorized access

ii) Education:

1. Ransomware
2. Phishing
3. Unauthorized access

iii) Finance:

1. Phishing
2. Ransomware
3. Unauthorized access

iv) Local government

1. Ransomware
2. Phishing
3. Malware

v) Legal

1. Phishing

2. Ransomware
3. Unauthorized access

c) What are the 3 main security incidents reported in the Health Sector and the finance?

- Three main security incidents reported in the Health Sectors and finance are:
 1. Data Emailed to wrong recipient
 2. Unauthorized access
 3. Data posted or faxed to incorrect recipients

d) What could be the reason that the incidents are higher in the health sector compared to the others?

- Some reasons are:
 1. Health care falls on critical infrastructure hackers can benefit to demand ransome
 2. Health sectors work utilizes more on personal records of the individuals

e) How have the reported incidents changed over the last 2 years?

- The incident of data sent to incorrect recipients remains on top with 1% increase in the last 2 years. Ransomware incidents has also got growth by 1% . Unauthorized access has also got growth of 1%. Theft of paperwork ot data left in insecure location has got decrease in 1%

Task 2

a. What percentage of UK businesses identified a cyber-attack or breach in the last 12 months?

- 50 % of UK business identified a cyber-attack or breach in the last 12 months.

b. What is the most common type of cyber-attack reported?

- The most common type of cyber-attack reported is Phishing.
- c. What percentage of businesses have a board member responsible for cyber security?
 - 30%
- d. How do micro and large businesses differ in their cyber security approaches?
 - Micro businesses cannot afford cybersecurity expertise than large businesses thus they has very low risk assessments and have informal response to incidents.
 - Large Business has professional cyber security expertise and formal response to the incidents
- e. What are the main reasons businesses do not invest more in cyber security?
 - The main reasons are:
 1. Affordability
 2. Lack of awareness
 3. Nature of Business
- f. What are the key components of the “10 Steps to Cyber Security” framework?
 1. Risk management
 2. Engagement and training
 3. Asset management
 4. Architecture and configuration
 5. Vulnerability management
 6. Identity and access management
 7. Data security
 8. Logging and monitoring
 9. Incident management
 10. Supply chain security
 -
- g. What percentage of businesses provide staff with regular cyber security training?
 - 18%
- h. How has remote working impacted cyber security risks?
 1. Via Exploitation of home network
 2. Data Leak
 3. Implementation of VPN networks

4. Jobs related to firewall and system monitoring has seen drastical increase overtime
- i. What challenges do businesses face when improving cyber resilience?
 1. In smaller business organizations are depended on ISP and DSP for cyber incidents
 2. Larger business has communication gaps between IT and Cyber Team for addressing cyber incidents
 - j. How many businesses have incident response plans in place and when do you need one?
 - 22% businesses have incident response plans in place. Incident response plans is needed before cyber security incident happens
 - k. What types of online services are most commonly used by businesses?
 1. ISP
 2. DSP
 3. Cloud Service Provider
 4. Payment Services
 - l. What is the relationship between BYOD (Bring Your Own Device) policies and businesses with online services?
 - Confidentiality
 - Integrity of data
 - External network cannot tamper with data
 - Implementation of VPN
 - m. On what date did General Data Protection Regulation (GDPR) supersede the Data Protection Act 1998 in the UK?
 - May 25, 2018
 - n. Do all firms in the UK consider cyber security a key priority?
 - No
 - o. What are the key reasons given by businesses for deprioritising cyber security?
 - Nature of Business
 - Affordability
 - Lack of Awareness

- p. Do all core staff consider cyber security important? Do some feel it hinders their daily activities?
- Not all core staff consider cyber security important. Yes some do hinder cyber security in their daily activities.
- q. What are the main reasons organisations invest in cyber security?
- Data protection
 - Legal compliance
 - Managing disruption
 - Preparedness for new threats
- r. Do all businesses have formal policies or document cyber security risks?
- No
- s. List some of the controls that businesses have implemented.
- Restricting admin rights
 - Network Firewalls
 - Malware Protection
 - Awareness of Phishing
- t. What kind of impacts can a business face when targeted by a cyber attack?
- Data Theft
 - Financial loss
 - Loss of reputation
 - Disruption in regular operations

Task 3

- a. What is the difference between risk, threat and vulnerability?]
- Risk in cyber security term is defined as the impact or incident after a flaw is exploited by the threats. Threats are the possible vectors that can exploit vulnerability of a system. Vulnerability is the flaw or error in the system which can be exploited by the possible threats.
- b. What is the difference between commodity and bespoke capability?

- Commodity are the tools and techniques that are readily available on the internet which are used for common purpose. They are publicly available and simple to use. Bespoke on the other hand is a tool or technique that is made for a specific purpose. It requires expertise to use.
- c. What is the difference between targeted and un-targeted attacks? Give some examples.
- Targeted attacks are those attacks which are done with specific intention to harm a chosen entity. A common example is spear phishing. Un-targeted attacks are those attacks which unintentionally trap an entity without being in the specific intention of the attacker. A common example are phishing, ransomware.
- d. When dealing with vulnerabilities, what is the difference between flaws, features and errors?
- Flaws are the unaware features that can be exploited by the threats.
 - Features are the legitimate service in a software or system which can be used to compromise a system by hacker
 - Error is a fault in a code of the system which interrupts the normal flow of system.
- e. What are the key stages of a cyber attack described by the Cyber Kill chain.
1. Survey
 2. Delivery
 3. Breach
 4. Affect
- f. How can the affects of each stage be mitigated?
- Malware Protection
 - Maintaining firewalls
 - Network defences

a. Read the Case Study 2 and Case Study 3

i. Compare the vulnerabilities exploited in each case

Vulnerabilities of Case study 2 are:

- a) Poor security Practices.
- b) Compromisation of the Internal network.

Vulnerabilities of Case study 3 are:

- a). Untrained Staff
- b) Internal Security Compromisation and flaws.

ii. Compare the mitigation steps in each case

For Case Study 2:

- a) By providing better security
- b) By securing the internal network.

For Case Study 3 :

- a) Training of staff and minimizing miscommunication.
- b) Better firewalls configurations and patching of any security vulnerabilities.